



Fig. 3: A nanogenerator consisting of an array of zinc oxide nanowires

able bonding strength between the nanowires and substrate as well as optimising the wire spacing is important in increasing the discharge efficiency. All the nanowires must generate electricity simultaneously and continuously, and all the electricity must be effectively collected and distributed.

Ceramic or semiconductor substrates that were used initially for growing zinc-oxide nanowires were hard and brittle, making them unsuitable for applications that require foldable or flexible power source, such as biosensors implanted in the human body or power generators built into shoes. It has been discovered that there are many polymers that are biocompatible and could be used effectively as a substrate. They are also conductive. Because of the flexibility of the substrate, the nanowire surface profile is wavy, causing some missed contacts. This problem can be overcome by providing suitable bonding strength between the nanowires and substrate as well as optimising the wire spacing, which increases the discharge efficiency. To improve adhesion of the nanowires with the substrate, a thin layer of polymer is added onto the substrate after growth so that the roots of the nanowires are partially embedded.

From about 6mm² size of a nanogenerator, presently 12 millivolts and 820 nanoamperes of electrical output can be available. Nanogenerators can be arranged in series to increase the output voltage and in parallel to improve the output current. Although the available power is quite small, these

are ideal generators for devices that need to work intermittently, such as sensors that collect and transmit for one second of every minute.

Blood flow or muscle movement squeezes and bends the piezoelectric zinc-oxide nanowires, generating an electric current. Because the wires are very thin—just 40 nanometres in diameter—these are very flexible and bend easily

to produce a comparatively large current.

Thermoelectric generator

Another option is nanothermoelectric generator based on Seebeck effect. Thermoelectricity is a phenomenon where heat energy is directly converted into electricity.

Thomas Seebeck discovered in 1821 that an electromotive force is produced by heating a junction between two metals. He found that near a closed circuit composed of two linear conductors of different metals, a magnetic needle would deflect if the two junctions were at different temperatures, and further that if the cooler junction were to become warmer, the direction of deflection would be reversed. The Seebeck EMF depends upon only the material and temperature. The effect he observed is the basis for thermoelectric power generation.

Thermoelectric effect can be best understood in terms of the motion of electrons and holes. Materials that are intrinsic semiconductors, are characterised by an electrical conductivity that is appreciably lower than of metals and rises rapidly with temperature. The inner electrons of an atom are closely bound to the nuclei and do not interact with other atoms. The outermost electrons of the atom are called valence electrons. The electron energy levels fall into two bands: a valence band of lower energy, which contributes to the bonding of crystals, and a conduction band of higher energy. The valence band is completely filled up with electrons and the conduction band is

empty. The two bands are separated by an energy gap.

In semiconductors, the separation between conduction and valence bands is quite small. Thus if a valence-band electron absorbs enough thermal energy, it will jump to the conduction band, becoming a free charge carrier. This event leaves a vacancy in the valence band, into which another electron can move. Thus movement of a single electron creates mobile charge in both the bands.

An n-type semiconductor material is made by adding a suitable type of impurity in small amount that introduces an excess of electrons in the crystalline structure. Impurity that produces a deficiency of electrons, i.e., holes, makes p-type semiconductors. These holes or excess electrons are termed as carriers.

The Seebeck coefficient, i.e., Seebeck voltage per unit of temperature, is small in most metals—typically a few microvolts per degree Celsius. Semiconductors have much larger values of coefficient, in the range of hundreds or thousands of microvolts per degree Celsius. Materials' properties required for thermoelectric application show what is needed is a semiconductor that is a good electrical conductor. Otherwise, electron scattering will generate heat on both sides of the barrier and throughout the material. Also, the materials must be poor thermal conductors, failing which the temperature difference that must be maintained between the hot and cold sides will produce large heat backflow.

In metals, most of the heat is carried by electrons or holes, which, at the same time, carry electric current. In any solid, heat is also transferred from hot to cold regions through vibrations of the atoms about their normal position in the crystal structure or lattice. This lattice thermal conductivity is the most important mechanism in semiconductors having small number of electrons or holes—less than one per 10,000 atoms.

The limit of performance of a